

COMPASS Synoptic CB Porewater: DOC

April 2023 Samples

2026-06-02

##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

Run information - PLEASE CHANGE

Date_Run = "04/30/23" *#Date that instrument was run*

Run_by = "Stephanie J. Wilson" *#Instrument user*

Script_run_by = "Stephanie J. Wilson" *#Code user*

run_notes = "Standard Curve was assessed manually on the instrument."

Check GCW SW samples, because the npoc values are 0.0000.

Some sample IDs are missing from metadata: 202304_MSM_UP_LysA_45cm, 202304_MSM_UP_LysB_45cm, 202304_MSM_UP_LysC_45cm, 202304_MSM_TR_LysA_45cm, 202304_MSM_TR_LysC_10cm, 202304_MSM_TR_LysC_20cm" *#any notes from the run*

samples <- c("GCW", "GWI", "MSM", "SWH") *#whatever identifies your samples within the same names*

samples_pattern <- paste(samples, collapse = "|")

#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code

chks_name = "3ppmN_50ppmC_Chk" *#what did you name your check standards?*

File Names - PLEASE CHANGE

#file path and name for raw summary data file

#raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505.txt" #example

raw_file_name = "Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202304.txt"

#file path and name for raw all peaks file

#raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt" #example

raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202411_allpeaks.txt"

#file path and name of processed data file

#processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202505.csv" #example

processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202304.csv" *#example*

Log Files - PLEASE CHECK

#downloaded metadata csv - downloaded from Google drive as csv for this year

Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2023.csv"

#qaqc log file path for this year

#Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAQClog_2024.csv"

```
##Set Up Code
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

Import Data Functions

Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
```

```
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>   <dbl> <chr>
## 1 202304_GCW_UP_LysB_10cm    14.6   0.459 5/2/2023 12:39:24 AM
## 2 202304_GCW_UP_LysB_20cm    11.9   0.590 5/2/2023 1:04:22 AM
## 3 202304_GCW_TR_LysA_20cm    38.2   1.52  5/2/2023 1:22:18 AM
## 4 202304_GCW_TR_LysB_10cm    15.7   0.482 5/2/2023 1:39:47 AM
## 5 202304_GCW_TR_LysB_20cm    27.6   1.33  5/2/2023 2:01:21 AM
## 6 202304_GCW_TR_LysB_45cm    50.4   1.06  5/2/2023 2:28:24 AM
```

Assessing Standard Curves - assessed manually on the instrument

Assess Check Standards

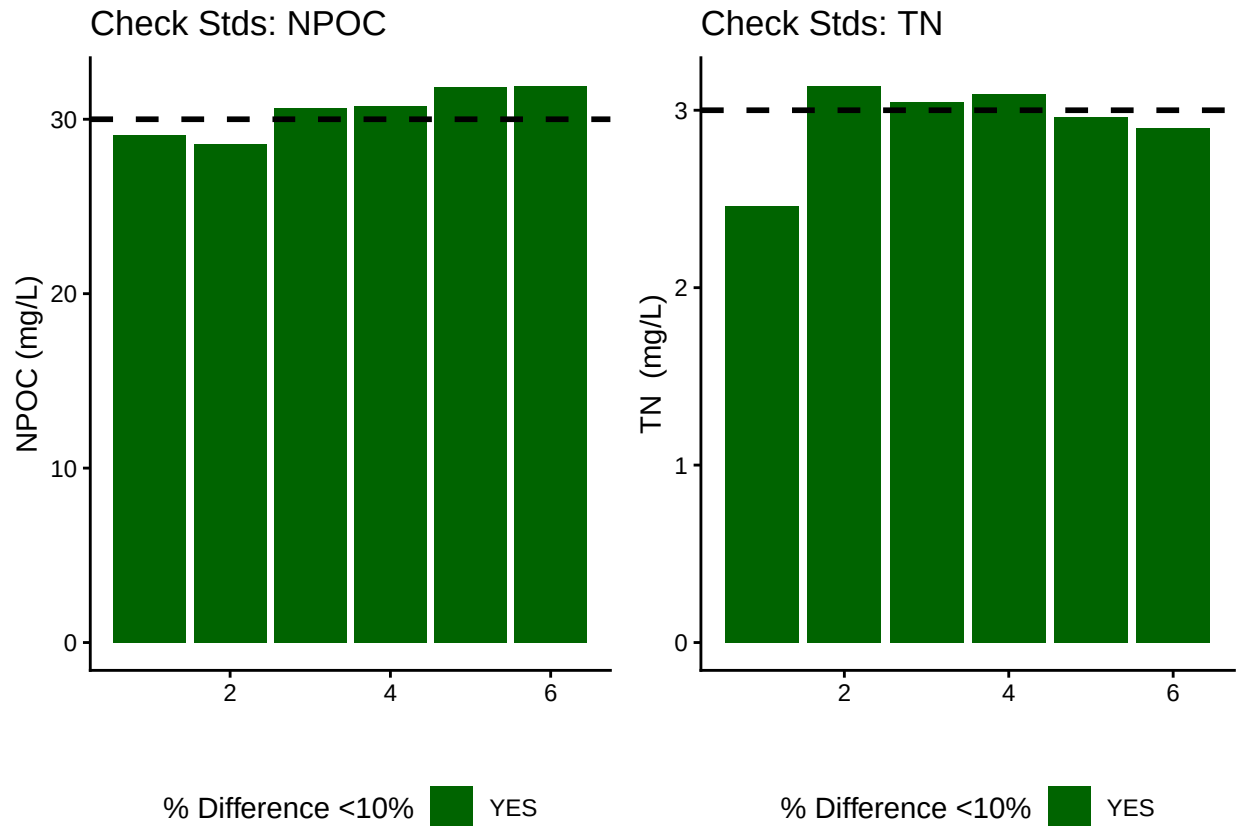
```
## Assess the Check Standards
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] "Carbon Check Standard RSD within Range"
```

```
## [1] "Nitrogen Check Standard RSD within Range"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

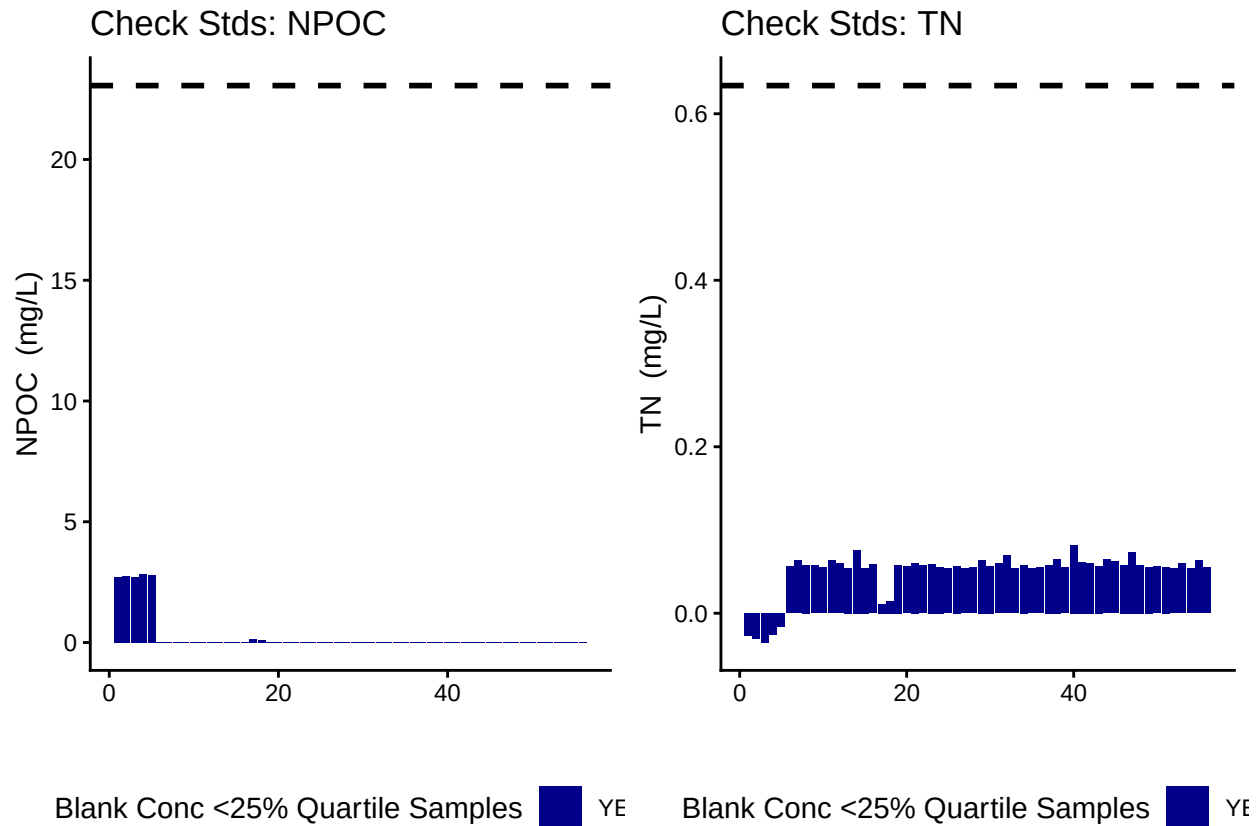
Assess Blanks

```
## Assess Blanks
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] 0.2516768
```

```
## nitrogen blanks:
```

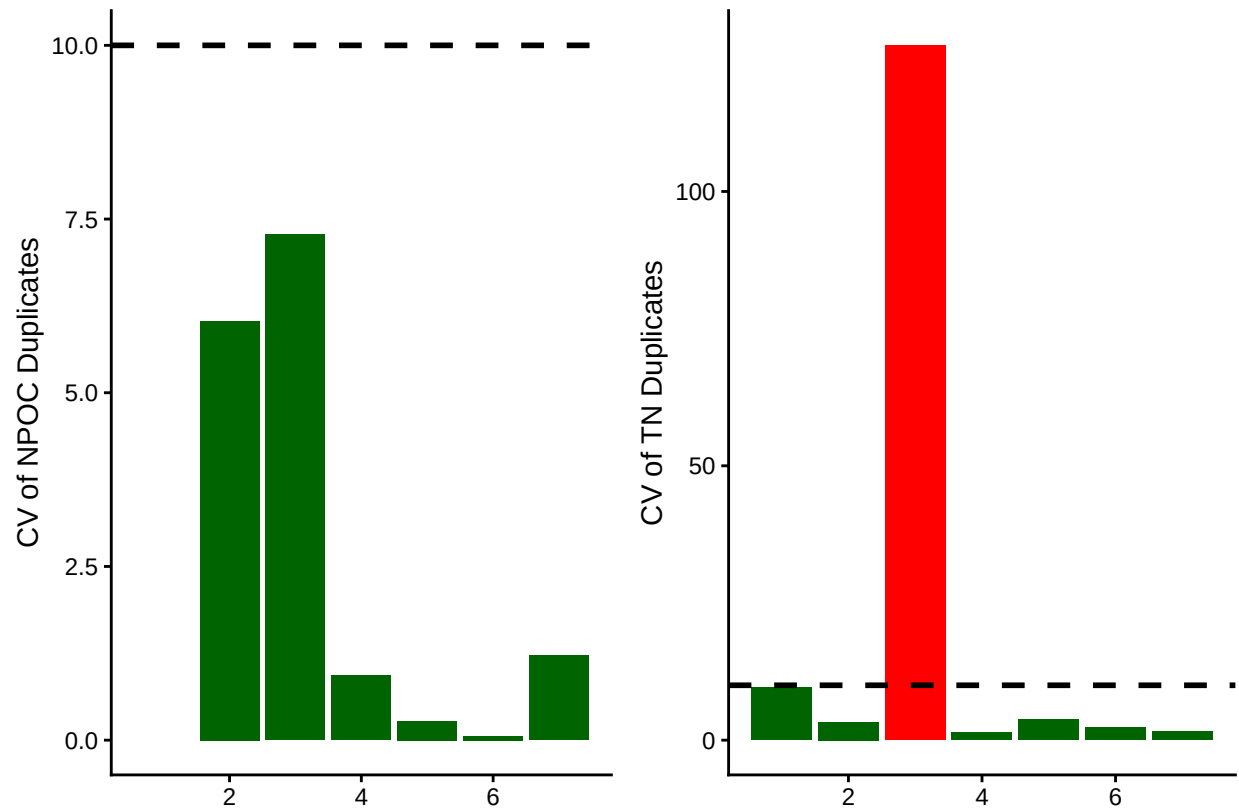
```
## [1] 0.04980786
```

Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```

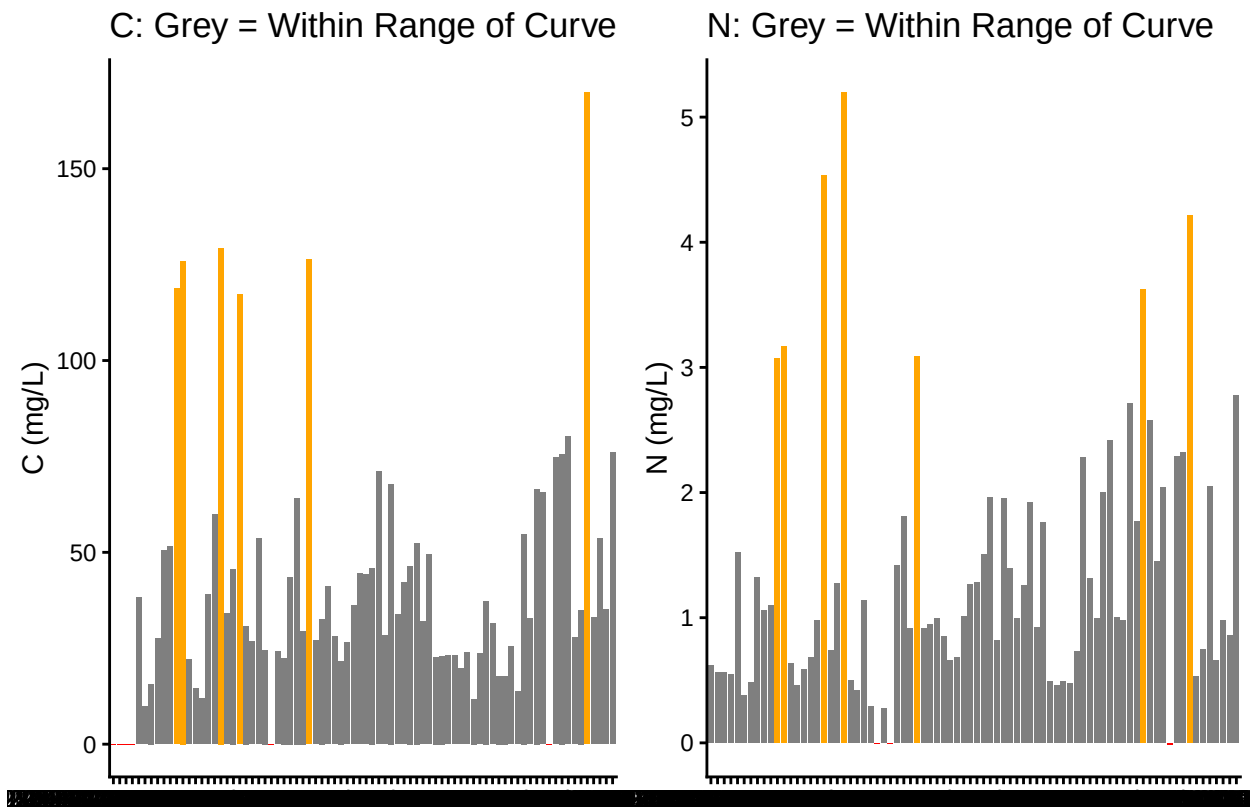


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

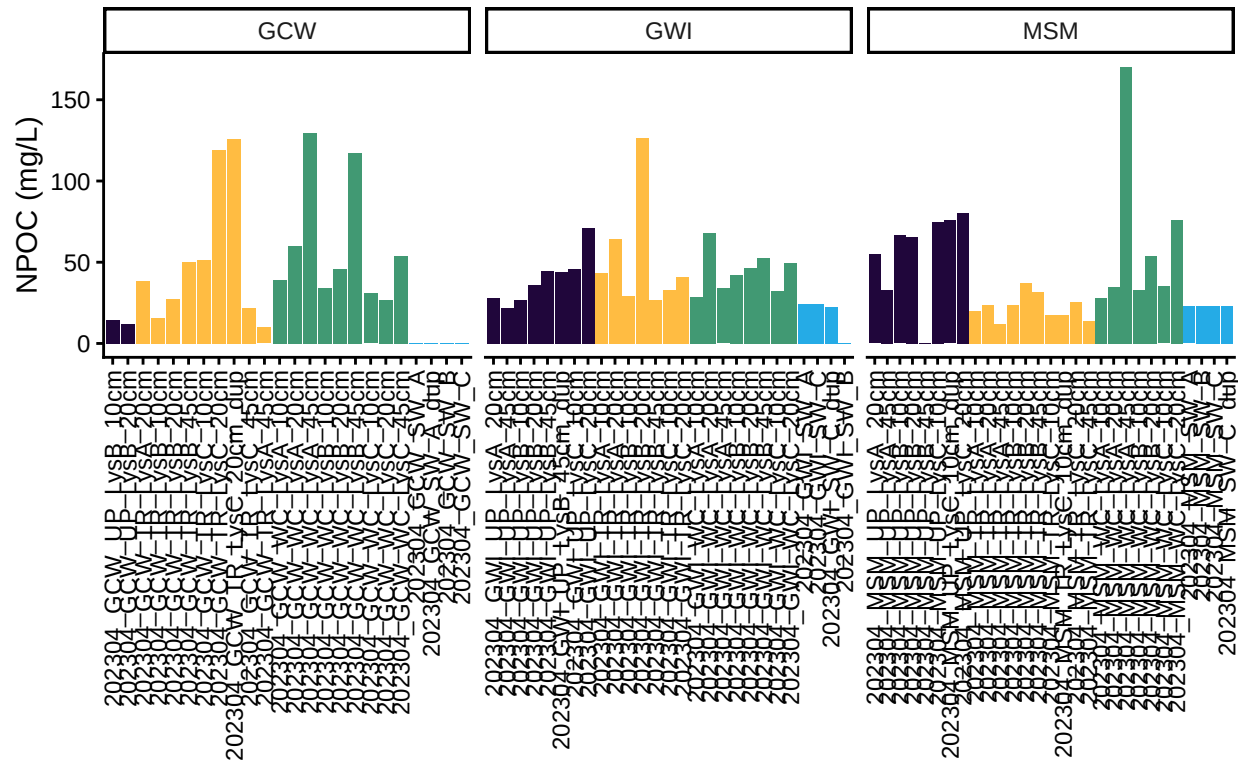


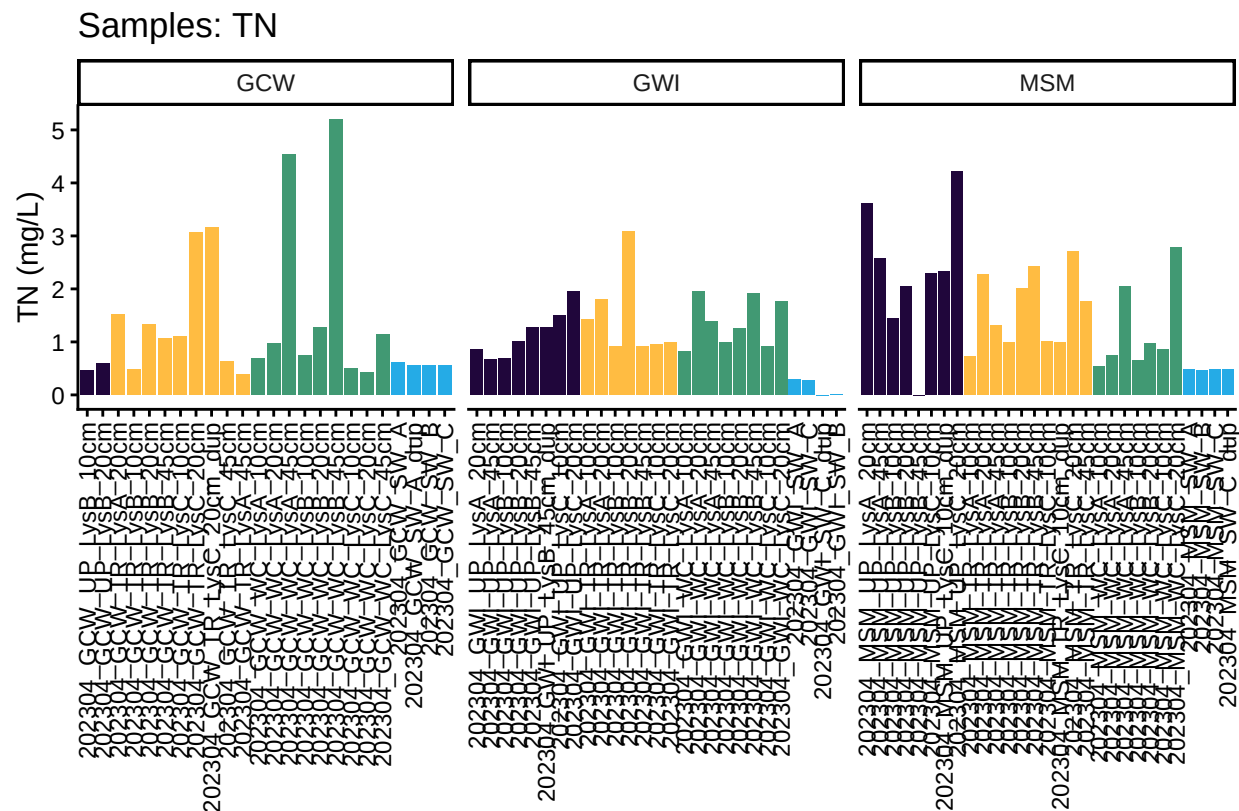
Visualize Data by Plot

```
## Visualize Data
```

```
## Warning in rbind(c("202304", "GCW", "UP", "LysB", "10cm"), c("202304", "GCW", :
## number of columns of result is not a multiple of vector length (arg 1)
```

Samples: NPOC





Convert data from mg/L to uMoles/L

Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata

## Some sample IDs are missing from metadata.

## [1] "202304_MSM_UP_LysA_45cm" "202304_MSM_UP_LysB_45cm"
## [3] "202304_GWI_UP_LysA_20cm" "202304_MSM_TR_LysA_45cm"
## [5] "202304_MSM_TR_LysC_10cm" "202304_MSM_TR_LysC_20cm"
```

Export Processed Data

```
## Export Processed Data

## # A tibble: 6 x 21
##   Project      Region Site  Zone Replicate Depth_cm Sample_ID  Year Month  Day
##   <chr>        <chr> <chr> <fct> <chr>      <int> <chr>    <int> <int> <int>
## 1 COMPASS: Sy~ CB    GCW  UP    B          10 202304_G~ 2023    4    19
## 2 COMPASS: Sy~ CB    GCW  UP    B          20 202304_G~ 2023    4    19
## 3 COMPASS: Sy~ CB    MSM  UP    A          20 202304_M~ 2023    4    24
## 4 COMPASS: Sy~ CB    MSM  UP    <NA>       NA 202304_M~   NA    NA    NA
```



```
## 5 COMPASS: Sy~ CB      MSM  UP   B              10 202304_M~ 2023      4      24
## 6 COMPASS: Sy~ CB      MSM  UP   B              20 202304_M~ 2023      4      24
## # i 11 more variables: Time <lgl>, Time_Zone <lgl>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

```
#end
```